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Modeling Nonlinear Insulin-Glucose Dynamics in Critically-ill Patients

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Abstract

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Keywords

Canvas Learning Management System, Docker containers, Performance tuning

Sammanfattning

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Nyckelord

Canvas Lärplattform, Dockerbehållare, Prestandajustering

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Chapter 1

Introduction

Patients in the intensive care unit suffering from critical illnesses (e.g. acute myocardial infarction, sepsis, stroke) have been widely documented to undergo hyperglycaemia, even those without a prior diagnosis of diabetes [1]. Acute stress observed in critical illnesses often induces physiological changes, as the secretion of cortisol and catecholamines results in an inflammatory response associated with insulin-resistance [2] [3] [4] [5] [6] [7]. While there is inconclusive evidence pertaining to the causality of stress hyperglycaemia and adverse outcomes, the onset of mortality is far greater in patients with no reported history of hyperglycaemia, in contrast to patients with diabetes [1] [3]. Sustaining glucose concentration under 6.106 mmol/L through insulin infusion has demonstrated a reduction of mortality and morbidity, compared to less-intensive methods of glycaemic control [8], at the subsequent cost of an increase of reported incidences of hypoglycaemia [9]. Further review of intensive insulin therapy and tight glucose control has identified discrepancies in the administration of glycaemic control, mostly corresponding to the risk of moderate to severe hypoglycaemia and glycaemic variability in a critically ill patient is equally detrimental as stress hyperglycaemia, with an increase of mortality and health complications observed [9] [10] [11]. Clinicians aim to maintain blood glucose in a range of 6 mmol/L to 10 mmol/L to account for adverse outcomes in hyperglycaemia and hypoglycaemia. Traditional intensive insulin therapy struggles to fully capture the dynamic fluctuations of blood glucose in critically ill patients. Thus, model-based methods are required to achieve tight glycaemic control.

1.1 Background

The implementation of model-based tight glucose control methods relies on the derivation of insulin-glucose dynamics. Mechanistic insulin-glucose models have been developed with an extensive physiological foundation, and have been validated for gluco-regulation in clinical settings [12] [13] [14] [15] [16]. The insulin-glucose models derived are nonlinear, which requires more advanced numerical methods to properly estimate the current and future insulin-glucose dynamics in patients. In practice, the estimated dynamics from the insulin-glucose models contain process noise from unaccounted dynamics in the model formulation as well as noisy measurements. As a result, the insulin-glucose dynamics in practice struggle to be estimated deterministically. To accommodate modeling of stochastic processes, statistical methods are needed to optimally estimate the dynamics in mechanistic insulin-glucose models.

1.2 Problem

1.2.1 Original problem and definition

There currently exist gaps in research for in-depth analysis of statistical modeling of insulin-glucose dynamics. Previous studies have employed Bayesian filtering methods for this task [17] [18], but they lack necessary evaluation and motivation regarding the patient cohort and the filter design process. While patient inputs differ among institutions due to differing nutrition guidelines and stay times for patients, a systematic virtual patient cohort framework would greatly improve the simulation of diverse patient dynamics and experiment reproducibility, as real patient data are often limited in size and closed-source. Concurrently, concise parameterization and optimization of Bayesian filters will further improve glycaemic control in critically-ill patients.

1.2.2 Scientific and engineering issues

To model the behavior of dynamical systems, experience of statistical modeling is often required to construct assumptions of the system parameters. In the case of insulin-glucose dynamics, there are several unknown patient-specific parameters, and each has its respective role in modeling gluco-regulation. As a result, on top of estimating nonlinear and time-varying dynamics, the models must be robust to misalignments of the assigned patient-specific parameters, which poses an ongoing challenge.

1.3 Purpose

Mechanistic insulin-glucose models are of great benefit to critically ill patients and clinicians, as modeling gluco-regulation allows for necessary glycaemic adjustments to avoid the onset of adverse outcomes. In practice, clinicians only have the ability to base insulin resistance on blood glucose measurements, and thus, state estimators models fill in the gap as a tool for insulin-glucose dynamics estimation. Improvements in the estimation of insulin-glucose dynamics can better capture the evolution of the health of the patient in the intensive care unit to prevent mortality and morbidity.

1.4 Goals

The overall goals of this project has been divided into the following sub-goals:

1. Design a virtual patient cohort
2. Compare performance of varying filtering configurations in estimation of insulin-glucose dynamics
3. Extrapolate the formulation of insulin-glucose estimators to estimate unknown patient-specific parameters

1.5 Research Methodology

The modeling of the insulin-glucose dynamics found in critically ill patients will be implemented on a virtual patient cohort. Using virtual patients allows the researcher to manually control the patient-specific parameters, which directly improves the evaluation of the outcome of the insulin-glucose control. Previous work will serve as the benchmark to compare how the parameterization of our method performs in estimating insulin-glucose dynamics of virtual patients. The chosen filter design will then be used to estimate additional dynamics not mathematically represented in insulin-glucose models. Fitting metrics and computational complexity of the estimators will be utilized to draw inferences about the performance.

1.6 Delimitations

This study will consist of a virtual patient cohort. Using virtual patients is important for maintaining internal validity in the experimental process, as it verifies the observed outcome of the experiment is due to the manipulation of patient-specific parameters rather than uncontrolled variables. The expense of using virtual patients can be identified in the reduced generalizability of this experiment, since virtual patients are models. While this is a limitation, this study will focus on the foundation of the modeling of insulin-glucose dynamics, and future work can replicate the study with real clinical data.

1.7 Structure of the thesis

Chapter 2 will introduce the existing modeling of insulin-glucose dynamics used, followed by an overview of discretization, state estimation, and related work in parameter estimation. In Chapter 3, the proposed research process used to answer the research questions of this study will be documented and motivated. The results will be presented in Chapter 4. The analysis of the results, key findings, and future work of the experiment will be summarized in Chapter **??**.

Chapter 2

Background

The Intensive Control Insulin-Nutrition-Glucose model presented by Lin et al. [14] contains the insulin-glucose dynamics in critically ill patients. The model is constructed as a fusion of two existing models validated for patients in the intensive care unit [12] [13], containing the following ensemble of differential equations

$$\dot{G} = -p_G G(t) - S_I(t)G(t)\frac{Q(t)}{1+\alpha_G Q(t)} + \frac{P(t) + E_G - C_G}{V_G} \tag{2.1}$$

$$\dot{Q} = n_I(I(t) - Q(t)) - n_C\frac{Q(t)}{1+\alpha_G Q(t)} \tag{2.2}$$

$$\begin{aligned}\dot{I} = &-n_K I(t) - n_L\frac{I(t)}{1+\alpha_I I(t)} - n_I(I(t) - Q(t)) + \frac{u_{ex}(t)}{V_I}\\ &+ \frac{(1-x_L)u_{en}(t)}{V_I}\end{aligned} \tag{2.3}$$

$$\dot{P}1 = -d_1 P1 + D(t) \tag{2.4}$$

$$\dot{P}2 = -\min\left(d_2 P2, P_{max}\right) + d_1 P1 \tag{2.5}$$

$$u_{en}(t) = k_1 \exp\left(-\frac{k_2}{k_3} I(t)\right), \quad \text{when C-peptide data is not available} \tag{2.6}$$

$$P(t) = \min\left(d_2 P2, P_{max}\right) + P_N(t). \tag{2.7}$$

Equation (2.1) is the independent insulin glucose removal, with G being the total blood glucose concentration (mmol/L). The whole-body insulin sensitivity $S_I(t)$ gives insight into the metabolic condition of the patient and is a time-varying dynamic. The insulin pharamacodynamics are shown in Equations (2.2) and (2.3), with Q and I being the interstitial insulin concentration (mU/L) and the plasma insulin concentration (mU/L) respectively. The exogenous insulin input is $u_{ex}(t)$ (mU/min), and the endogenous insulin production $u_{en}(t)$ (mU/min) can be measured from C-Peptide, but if measurements are not feasible, the kinetics can model as such in Equation (2.6). The gastric absorption of glucose is defined in Equations (2.4) to (2.7). The dynamics $P1$ and $P2$ describe the glucose in the stomach and the gut (mmol), with $P(t)$ representing the glucose appearance in the blood (mmol/min). Any enteral food intake given to the patient is accounted for by $D(t)$ (mmol/min), and additional parenteral glucose is described by $P_N(t)$ (mmol/min).

The Intensive Control Insulin-Nutrition-Glucose model contains of several fixed parameters detailed in Table 2.1. The majority of the patient model parameters are assigned as general population constants observed from clinical trials, and are validated in relevant insulin-glucose control parameter sensitivity studies [15] [16] [19]. The identification of the remaining constant parameters was estimated with a data-driven approach by fitting the patient parameters to the observed dynamics in 173 patients, and by assessing the impact of prediction errors [19]. Estimating the endogenous glucose removal p_G and the suppression of endogenous glucose produce E_G constants involved assigning all other patient parameters and the insulin sensitivity to constants, then starting a grid search that covers $p_G = 0.001, 0.002, \ldots, 0.1(\text{min}^{-1})$ and $E_G = 0.0, 0.1, \ldots, 3.5$ (mmol/min). The joint p_G and E_G value that achieved the lowest fitting and prediction error for blood glucose in Equation (2.1) was chosen. Ultimately, that was $p_G = 0.006(\text{min}^{-1})$ and $E_G = 1.16$, while both falling in the ranges observed physiologically. The glucose distribution rate between the stomach and the gut n_I, and cellular insulin clearance rate in the interstitium were estimated similarly by fitting to Q in the Equation (2.2), a grid search covers a range of $n_I = n_C = 0.0001, \ldots, 0.02(\text{min}^{-1})$, with a value of $n_I = 0.003$ resulting in the best predictive performance.

It should be noted that in the Intensive Control Insulin-Nutrition-Glucose model, the nonlinear dynamics are defined in continuous time. Discretization of the dynamic system is necessary as in the following section, the estimation methods presented operate within discrete time, noted as k. If we define the discrete time step as $t_k = t_{k-1} + dt$, an example dynamic $F(t)$ can be discretized according to the Euler method

$$\begin{aligned} \dot{F} &= \frac{F(t_k) - F(t_k - dt)}{dt} \\ F_k &= F_{k-1} + dt\dot{F}. \end{aligned} \tag{2.8}$$

The error of this numerical approximation of differential equations is bounded

Table 2.1: Constant parameters in the Intensive Control Insulin-Nutrition-Glucose model.

Constant Parameter	**Notation**	**Value**
Patient endogenous glucose removal	p_G (min^{-1})	0.006
Saturation of insulin-stimulated glucose removal	α_G (L/mU)	0.0154
Suppression of endogenous glucose production	E_G (mmol/min)	1.16
Central nervous system uptake	C_G (mmol/L)	0.3
Glucose distribution volume	V_G (L)	13.3
Glucose distribution rate between the stomach and the gut	n_I (min^{-1})	0.003
Cellular insulin clearance rate in the interstitium	n_C (min^{-1})	$= n_I$
Kidney clearance rate of insulin from plasma	n_K (min^{-1})	0.0542
Liver clearance rate of insulin from plasma	n_L (min^{-1})	0.01578
Saturation of plasma insulin disappearance	α_I (L/mU)	0.0017
Insulin distribution volume	V_I (L)	3.15
Fraction of endogenous insulin hepatic extraction	x_L	0.67
Maximal glucose out flux from the gut	P_{max} (mmol/min)	6.11
Base endogenous insulin production rate	k_1 (mU/min)	45.7
Transport rates between the stomach and the gut	d_1 (min^{-1})	0.0347
	d_2 (min^{-1})	0.0069
Exponential suppression constants	k_2	1.5
	k_3	1000

Parameters shaded in gray are derived from fitting methods, as the rest are assigned as population constants.

by the step size dt. As a result, larger step sizes will introduce greater approximation errors. Thus, a relatively small step size is necessary to prevent errors from propagating in the estimates.

2.1 State Estimation

Within the scope of this paper, discrete-time dynamical systems will be modeled in the state-space representation. A state-space model consists of system variables that describe the state of the system, inputs, and measurements. The states are typically formulated as linear or nonlinear differential equations. A probabilistic representation of the discrete-time state-space can be represented as such

$$\begin{aligned} x_k &\sim p(x_k \mid x_{k-1}, u_{k-1}) \\ y_k &\sim p(y_k \mid x_k), \end{aligned} \tag{2.9}$$

where $p(x_k \mid x_{k-1},\, u_k)$ and $p(y_k \mid x_k)$ are the stochastic discrete-time dynamic and measurement models, with $x_k \in \mathbb{R}^n$ representing the states, $u_k \in \mathbb{R}^p$ are the inputs, and $y_k \in \mathbb{R}^m$ are the measurements.

Any dynamic model has intrinsic approximation errors, and the errors should be taken into account. The approximation error for the states can be modeled as a random process, and artificial process noise can be added into x_k at each time step. This step is important to accommodate errors due to unaccounted dynamics, and for notation purposes, the artificial process noise will be omitted from the probabilistic representation in Equation (2.9). Choices in process noise will be discussed in Chapter 3.

The objective of state estimators can described as estimating the distribution of the state x_k , given all the measurements up to the current sample $p(x_k \mid y_{1:k})$. In the case of assuming the state estimator is linear and Gaussian, the states and measurements can be written as the difference equations

$$\begin{aligned} x_k &= A_{k-1}x_{k-1} + B_{k-1}u_{k-1} + q_{k-1} \\ y_k &= Hx_k + r_k, \end{aligned} \tag{2.10}$$

where $q_{k-1} \sim \mathcal{N}(0,\, Q_{k-1})$ is the process noise, $r_k \sim \mathcal{N}(0,\, R_k)$ is the measurement noise, we can rewrite the Equation (2.9) as the following

$$\begin{aligned} p(x_k \mid x_{k-1}, u_{k-1}) &= \mathcal{N}(x_k \mid A_{k-1}x_{k-1} + B_{k-1}u_{k-1},\, Q_{k-1}) \\ p(y_k \mid x_k) &= \mathcal{N}(y_k \mid H_k x_k,\, R_k). \end{aligned} \tag{2.11}$$

The closed-form solution of $p(x_k \mid y_{1:k})$ for linear and Gaussian dynamic and measurement functions is known as the Kalman filter, with a full derivation available from Särkkä and Svensson [20]. The filter contains two key parts: the *prediction step* and the *update step*.

1. *Prediction step*

$$\begin{aligned} m_k^- &= A_{k-1}x_{k-1} + B_{k-1}u_{k-1} \\ P_k^- &= A_{k-1}P_{k-1}A_{k-1}^T + Q_{k-1}. \end{aligned} \tag{2.12}$$

2. *Update step*

$$\begin{aligned} v_k &= y_k - H_k m_k^- \\ S_k &= H_k P_k^- H_k^T + R_k \\ K_k &= P_k^- H_k^T S_k^{-1} \\ m_k &= m_k^- + K_k v_K \\ P_k &= P_k^- - K_k S_k K_k^T. \end{aligned} \tag{2.13}$$

The Kalman filter is recursive, as the filter starts with an estimate of the initial distribution of the state m_0^- and the covariance P_0^-, and iterates sequentially over discrete time steps k. In practice, the Kalman filter is often a great starting point for an estimator because the assumptions of linearity and Gaussianity often hold in many contexts, but as Equations (2.1) to (2.5) are nonlinear, other methods must be considered.

2.1.1 Extended Kalman filter

The extended Kalman filter can accommodate the nonlinearities in the Intensive Control Insulin-Nutrition-Glucose model by approximating the dynamic and measurement models as a linear transformation of the states and inputs. The structure of the extended Kalman filter remains the same as the Equations (2.12) and (2.13), but changes are seen in the matrices of the dynamic and measurement models $f(\cdot)$ and $h(\cdot)$. Assuming that the dynamic and measurement models are Gaussian, the prediction and update steps [20] can be defined as

1. *Prediction step*

$$\begin{aligned} m_k^- &= f(m_{k-1}, u_{k-1}) \\ P_k^- &= F_x(m_{k-1}) P_{k-1} F_x^T(m_{k-1}) + Q_{k-1} \end{aligned} \tag{2.14}$$

2. *Update step*

$$\begin{aligned} v_k &= y_k - h(m_k^-) \\ S_k &= H_x(m_k^-) P_k^- H_x^T(m_k^-) + R_k \\ K_k &= P_k^- H_k^T(m_k^-) S_k^{-1} \\ m_k &= m_k^- + K_k v_k \\ P_k &= P_k^- - K_k S_k K_k^T, \end{aligned} \tag{2.15}$$

the Jacobian matrices $F_x(m_k^-)$ and $H_x(m_k^-)$ are the Taylor series linear approximation of the dynamic and measurement models

$$\begin{aligned} [F_x(m)]_{j,j'} &= \left. \frac{\partial f_j(x)}{\partial x_{j'}} \right|_{x=m_{k-1}} \\ [H_x(m)]_{j,j'} &= \left. \frac{\partial h_j(x)}{\partial x_{j'}} \right|_{x=m_k^-} . \end{aligned} \tag{2.16}$$

The intrinsic bottleneck of the extended Kalman filter can be ascribed to the Gaussian formulation of the state update equations. Any deviations from Gaussianity in the posterior will directly degrade the estimation capabilities of the estimator. Another type of state estimation method is the particle filter, which can accommodate nonlinear and nongaussian dynamic and measurement models by using point estimates from a Monte Carlo approximation of the posterior $p(x_k \mid y_{1:k})$.

2.1.2 Bootstrap particle filter

For state-space modeling, the dynamic and measurement models are run sequentially such that x_k and y_k are represented as a Markov Chain

$$\begin{aligned} x_k &= p(x_k \mid x_{k-1},\, u_{k-1}) \\ y_k &= p(y_k \mid x_k) \end{aligned} \tag{2.17}$$

In the Kalman filter case, such dynamic and measurement models were assumed to be linear and Gaussian. As a result, the estimate of the state x_k was derived as a linear transformation of the mean and covariance. For the bootstrap particle filter, a point estimate of the state is formulated as a Monte Carlo approximation, with $x_k^{(i)}$ denoting a randomly sampled state for N total particles

$$\hat{x_k} = \frac{\sum_{i=1}^{N} p(y_k \mid x_k^{(i)}) x_k^{(i)}}{\sum_{j=1}^{N} p(y_k \mid x_k^{(j)})}. \tag{2.18}$$

The steps to implement to formulate the bootstrap particle fiter iteratively can be summarized in the following algorithm [20] [21] [22]

1. Sample N particles from the dynamic function

$$x_k^{(i)} = p(x_k \mid x_{k-1}^{(i)},\, u_{k-1}) \quad i = 1, \ldots, N \tag{2.19}$$

2. Update weights from the measurement

$$\begin{aligned} w_k^{*(i)} &\propto p(y_k \mid x_k^{(i)}) \quad i = 1, \ldots, N \\ w_k^{(i)} &= \frac{w_k^{*(i)}}{\sum_{j=1}^{N} w_k^{*(j)}} \quad i = 1, \ldots, N \end{aligned} \tag{2.20}$$

3. Systematically resample particles from a distribution proportional to the

weights

$$u_1 \sim \mathbb{U}\left[0, \frac{1}{N}\right] \tag{2.21a}$$

$$u_r = u_1 + \frac{r-1}{N} \qquad r = 2, \ldots, N \tag{2.21b}$$

$$x_k^{(i)} = \left\{ x_k^{(r)} \quad \text{if } \sum_{j=1}^{i-1} w_k^{(j)} < u_r \leq \sum_{j=1}^{i} w_k^{(j)} \right. \qquad i = 1, \ldots, N \tag{2.21c}$$

$$w_k^{(i)} = \frac{1}{N} \qquad i = 1, \ldots, N \tag{2.21d}$$

2.1.3 Observability in Nonlinear Discrete-Time State-Space Models

Analyzing observability in linear and nonlinear systems assesses how well the dynamics of the model can be inferred from the observations. The observability gramian W_O has traditionally been used in model reduction of linear dynamic systems, but empirical methods have been adapted to accommodate nonlinear systems [23]. The observability gramian provides insight which states have the most influence on the output of the system by analyzing the rank condition of the observability gramian matrix, which in return allows for analytically evaluating observability.

To empirically calculate the observability gramian, we first generate 2^n unique combinations of the initial state $x_0 \in \mathbb{R}^n$, with n denoting the numbers of states . The initial condition x_0 will be assigned as perturbations $\delta_i \in \mathbb{R}^n$ around a nominal value $x_{nom} \in \mathbb{R}^n$

$$\begin{aligned} x_{0,i} &= x_{nom} + \delta_i \\ \delta_i &= c \cdot S \cdot t_i, \qquad \text{with } S = \operatorname{diag}(x_{\text{max}}). \end{aligned} \tag{2.22}$$

The scalar constant c corresponds to the magnitude of deviation, $S \in \mathbb{R}^{n,n}$ is the diagonal matrix of the maximum of each state and parameter. The row vector $t_i \in R^n$ is the specified perturbation i from the orthonormal matrix

$T = \begin{bmatrix} t_1 & t_2 & \dots & t_{2^n} \end{bmatrix} \in R^{n,2^n}$, such that

$$\begin{aligned} t_1^T &= \frac{1}{\sqrt{2^n}} \begin{bmatrix} -1 & -1 & \dots & -1 \end{bmatrix} \\ t_2^T &= \frac{1}{\sqrt{2^n}} \begin{bmatrix} -1 & -1 & \dots & 1 \end{bmatrix} \\ &\vdots \\ t_{2^n}^T &= \frac{1}{\sqrt{2^n}} \begin{bmatrix} 1 & 1 & \dots & 1 \end{bmatrix}. \end{aligned} \tag{2.23}$$

After calculating the 2^n perturbations of the initial state, the output can be derived sequentially with Equations (2.9) from time step $k = 1, \dots, K$ beginning with $x_{0,i}$ to generate $y_i[1 : K]$. Then, the empirical observability with discrete time steps $t = k\,dt$ can be expressed as such

$$W_O = (cS)^{-1} \sum_{k=1}^{K} T\Psi[k]\,T^T\,dt(cS)^{-1}, \tag{2.24}$$

with the positive definite matrix $\Psi[k]$ being assigned as the following

$$\Psi_{i,j}[k] = |y_i[k] - y_0[k]|\,|y_j[k] - y_0[k]|\,. \tag{2.25}$$

If the states have varying magnitudes of scale, it is recommended [18] to rerun the observability gramian algorithm again with the following transformations

$$x_s = \sqrt{\operatorname{diag}(W_O)} \qquad\qquad T_s = \operatorname{diag}^{-1}(x_s). \tag{2.26}$$

The new state perturbations are now assigned as the transformation $\bar{x} = T_s x$. To make inferences about of the observability gramian, one can check the rank condition of the matrix by analyzing the nonzero singular values of W_0. If there exists singular values near zero (e.g. < 0.1),a subset of the dynamics of the system can not be observed from the measurements.

2.1.4 Related Work

Nonlinear state estimation of the Intensive Control Insulin-Nutrition-Glucose model has been investigated in a previous study by Aguirre-Zapata et al. [17].

The study incorporated an additional measurement and dynamic due to their clinical facility having a continuous glucose machine that can measure the patient's interstitial glucose, but the formulation of patient dynamics otherwise remained the same. In the methodology of the paper, all tests regarding estimation performance are tested on deviations from one virtual patient, with each deviation being assigned the same exogenous insulin input per deviation, and food intake is generally not considered. Overall, their virtual patients don't characterize a reasonable patient model in comparison to real patients.

The study includes three state estimators: the extended Kalman filter, the unscented Kalman filter, and the particle filter. The majority of the study uses discrete time-steps of 5 minutes. While the dynamics of patients develop relatively slowly, such a large time step can introduce approximation errors in the discretization of the ICING model. Another notable design choice of their particle filter is that it contains 5000 particles. Further analysis of various sized particle filters would motivate the overall trade-off estimation performance compared to the time complexity. State estimation remained in the scope of dynamics represented in the Intensive Control Insulin-Nutrition-Glucose model and an additional interstitial glucose dynamic. As a result, there exists further development of state estimators to predict unrepresented patient-specific parameters as well.

2.2 Parameter Estimation

As previously mentioned, there exists patient-specific parameters. The whole-body insulin sensitivity S_I or the endogenous insulin u_{en} dynamically fluctuates over time depending on the patient's metabolic condition. As a result, naively assigning either a population constant can hinder the performance of the estimators. The unknown constant parameters of interest will be denoted as θ.

2.2.1 State Augmentation

A practical method to handle unknown parameters is to augment them to the existing state variables x_k, such that Equation (2.9) can be redefined as

$$\begin{aligned} &\theta_k = \theta_{k-1} \\ &x_k \sim p(x_k \mid x_{k-1},\, u_{k-1},\, \theta_{k-1}) \\ &y_k \sim p(y_k \mid x_k). \end{aligned} \tag{2.27}$$

An important distinction in Equation (2.27) is that the unknown parameters are treated as stochastic variables with singularity in the process noise. This can be problematic as all models often have approximation errors, and this can drive the extended Kalman filter to ultimately diverge in the Gaussian approximation, as well as introduce sample impoverishment in particle filters [20].

One method to compensate for singularity is to also add artificial process noise Q_k to the unknown parameters to allow for random walking in the state space, thus, accounting for errors in the dynamic function

$$\theta_k = \theta_{k-1} + \epsilon_{k-1}, \qquad \text{with } \epsilon_{k-1} \sim \mathcal{N}(0,\, Q_{k-1}). \tag{2.28}$$

With this method, you inherently add another unknown parameter Q_k, but such a parameter can be tuned with prior knowledge of the parameters θ_k.

2.2.2 Related Work

A nonlinear, and nonconvex integral-based parameter identification method has been introduced that can estimate patient-specific parameters as a least squares solution [15]. By linearly interpolating between the blood glucose measurements in times $[t_0, t_1]$,

$$\widetilde{G(t)} = \frac{G(t_1) - G(t_0)}{t_1 - t_0} t, \tag{2.29}$$

the Equation (2.1) can be rearranged as such for a piecewise constant S_I

$$\begin{aligned}
\int_{t_0}^{t_1} \dot{G}\ dt &= -p_G \int_{t_0}^{t_1} \widetilde{G(t)}\ dt - S_I \int_{t_0}^{t_1} \widetilde{G(t)} \frac{Q(t)}{1 + \alpha_G Q(t)}\ dt \\
&\quad + \frac{1}{V_G} \int_{t_0}^{t_1} P(t) + E_G - C_G\ dt \\
G(t_1) - G(t_0) &= -p_G \int_{t_0}^{t_1} \widetilde{G(t)}\ dt - S_I \int_{t_0}^{t_1} \widetilde{G(t)} \frac{Q(t)}{1 + \alpha_G Q(t)}\ dt \\
&\quad + \frac{1}{V_G} \int_{t_0}^{t_1} P(t) + E_G - C_G\ dt,
\end{aligned} \tag{2.30}$$

and S_I can be expressed as a linear equation

$$\underbrace{\int_{t_0}^{t_1} \widetilde{G(t)} \frac{Q(t)}{1+\alpha_G Q(t)}\, dt}_{A(t_0,t_1)}\ S_I = \underbrace{\begin{pmatrix} G(t_0) - G(t_1) - p_G \displaystyle\int_{t_0}^{t_1} \widetilde{G}(t)\ dt \\ + \dfrac{1}{V_G} \displaystyle\int_{t_0}^{t_1} (P(t) + E_G - C_G)\ dt \end{pmatrix}}_{b(t_0,t_1)} . \tag{2.31}$$

It is often the case with longer measurement periods ($t_1 - t_0$), that identifying multiple S_I values within the time period may be necessary. For instance, if every 120 minutes glucose measurements are taken from the patient, but every 60 minutes a patient's insulin sensitivity should be identified $S_I \in \mathbb{R}^2$. Both insulin sensitivity parameters at 60 minutes S_I^{60} and 120 minutes S_I^{120} can be derived as the least squares solution from the system of linear equations

$$\begin{bmatrix} A(t_0, t_0+30) & 0 \\ A(t_0+30, t_0+60) & 0 \\ 0 & A(t_0+60, t_0+90) \\ 0 & A(t_0+90, t_0+120) \end{bmatrix} \begin{bmatrix} S_I^{60} \\ S_I^{120} \end{bmatrix} = \begin{bmatrix} b(t_0, t_0+30) \\ b(t_0+30, t_0+60) \\ b(t_0+60, t_0+90) \\ b(t_0+90, t_0+120) \end{bmatrix} . \tag{2.32}$$

The fundamental flaw of the integral-based parameter identification method is the assumption of linear blood glucose dynamics between measurements of times $[t_0, t_1]$. If we closely analyze the Figure 2.1, linearly interpolating blood glucose introduces residuals, due to fitting errors. Consequently, the residuals will propagate into the integrals in the Equation (2.31), resulting in errors in the parameter calculation. Relying on data-driven methods can not account for physiological behavior in the parameters, and other methods must be explored to allow for realistic estimation of parameters, thus giving better insight of the patient's health status to clinicians.

2.3 Summary

The Intensive Control Insulin-Nutrition-Glucose model provides a foundation to model glucose removal, insulin pharmacokinetics, and gastric absorption of glucose for tight glucose control in critically ill patients. Utilizing nonlinear state estimators such as the extended Kalman filter and the bootstrap particle filter allows for optimal estimation of the dynamics over time with consideration of model approximation errors and measurement

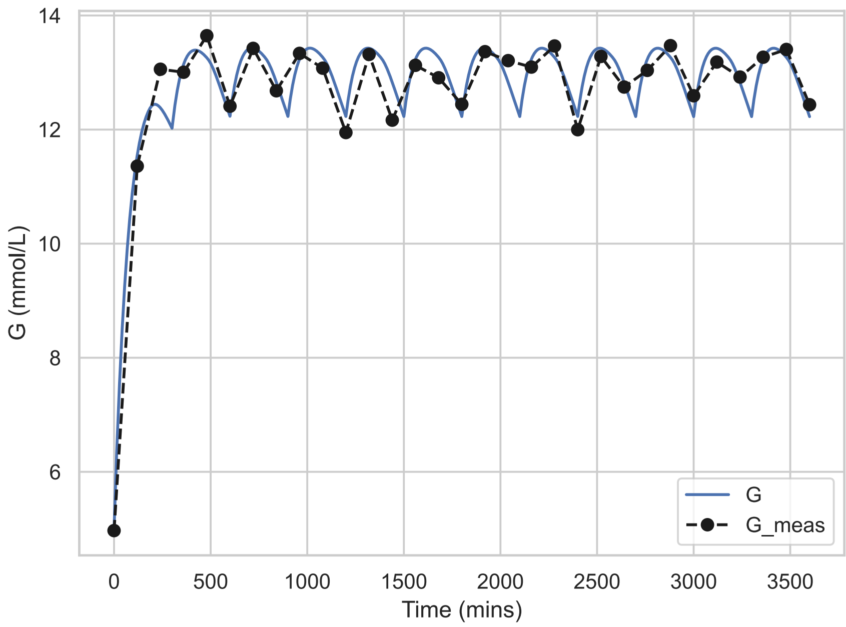


Figure 2.1: Blood glucose and its' corresponding measurements from an example patient model.

noise. Consideration of realistic virtual patients and systematically motivated parameterization of filters will improve generalizability of the estimators. The current method of integral-based parameter identification is formulated with strict assumptions of blood glucose measurements, which limits the ability for clinicians to infer the health of the patient from the approximated patient-specific parameters. More in-depth parameter estimation methods can improve model identifiability for glucoregulation in critically ill patients.

Chapter 3

Methodology

This chapter will consist of presenting the scientific design of the experiment. Beginning with the overall goals of the study, and how the corresponding experimental process will be formulated to achieve answering the stated research questions. The inclusion and motivation of the data collected, models used, and analysis of the observations in the experiment will be laid out in the following sections.

3.1 Research Process

The main topics this study aims to address fall in line with the following research questions:

1. **How do state estimation techniques compare in estimating patient insulin-glucose dynamics in terms of accuracy and computational complexity?**
2. **How do practical parameter estimation methods perform relative to the integral-based parameter identification method in terms of estimating unknown constant and time-varying patient-specific parameters?**

Given the overall experimental goals, the research process can therefore be outlined as such:

Step 1 create virtual patients,

Step 2 conduct experiments with patient data,

Step 3 analyze statistical metrics from the experiments, and

Step 4 discuss the results of the analysis.

3.2 Research Paradigm

The philosophical motivation in this study is founded on utilizing a set of quantitative methods that serve the purpose of ultimately answering the previously stated research questions. The use of statistical metrics will quantitatively express the relationships of variables tested in the experiment. Consequently, the research questions will effectively be addressed.

3.3 Data Generation

A virtual patient cohort will be simulated to model clinical insulin-glucose data observed in patients in the intensive care unit. A patient model consists of simulated insulin-glucose dynamics (G, Q, I, $P1$, $P2$, u_{en}, P) represented in Equations (2.1) to (2.7), known insulin sensitivity S_I, exogenous insulin u_{ex}, enteral feed D, and parenteral feed PN. The population constants listed in Table 2.1 are assigned to the constant parameters for all virtual patients. To introduce variance between patients, a virtual patient is given a random constant input observed from doses given to patients in the Karolinska University Hospital

$$\begin{aligned} u_{ex} &\sim \mathcal{U}[0,\, 166] \text{ mU/min} \\ D &\sim \mathcal{U}[0.2,\, 0.4] \text{ mmol/min} \\ PN &\sim \mathcal{U}[0.2,\, 0.4] \text{ mmol/min.} \end{aligned} \tag{3.1}$$

The insulin sensitivity SI has been shown to follow a log-normal distribution [24]. Sampling the insulin sensitivity from a log-normal is important to adhere to the physiological constraints typically observed in patients, as the insulin sensitivity is small in magnitude, and sampling from a normal distribution may shift the value to negative or too large in magnitude, which can possibly result in poor estimation. The values chosen within this study will be drawn from the distribution*

$$SI = \exp(X), \qquad \text{with } X \sim \mathcal{N}(-8.05,\, 0.70^2). \tag{3.2}$$

To simulate patient data given the inputs and parameters, each virtual

*The distribution was chosen as the natural logarithm (base e), instead of base 10 according to Davidson et .al [24]. Although the basis differs, the SI remains normally distributed in the base e domain.

patient needs to be assigned an unique initial state* for the dynamics

$$x_0 = \begin{bmatrix} G_0 & I_0 & Q_0 & P1_0 & P2_0 & u_{en0} \end{bmatrix}. \tag{3.3}$$

If it assumed that the patient model begins at a steady state, it is possible to derive the initial states with the given input and parameter values by solving for the roots of the Differential Equations (2.1) to (2.6). This can be computed with the solver function `scipy.optimize.fsolve`† for each patient.

Given all the initial states x_0, known insulin sensitivity S_I, and the patient inputs u_{ex}, D, and PN, the patient cohort can be generated by solving the Differential Equations (2.1) to (2.5) with a time step of $dt = 1$ min for each patient, and the results will be inputs to Equations (2.6) and (2.7). The ODE solver `scipy.integrate.solve_ivp`‡ function in python will be applied with the Explicit Runge-Kutta of order 5(4) integration method [25] to solve for the necessary dynamics.

3.4 Assessing reliability and validity of the data generated

3.4.1 Validity of method

The patient models, parameters, and initial states are all adopted and cross-validated from multiple clinically validated studies. This provides a physiologically accurate basis for the control variables of the experiment. Interventions in the experiment can be assessed and compared to the manipulated variables, and with confidence a conclusion can be reached that the intervention is correlated with the change in the output, and not due to poor model formulation with nonvalidated patient input and parameters.

3.4.2 Reliability of method

Simulating patient data gives the necessary control to assign all patient parameters. A clear benefit of virtual patients is the repeatability of

*The glucose appearance in the blood P is not included as an initial state in the state-space model. This is a result of patients not reaching the maximum glucose out flux from the gut P_{max} in the beginning of the situation, thus, it simplifies to a deterministic equation.

†https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.fsolve.html

‡https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.solve_ivp.html

experiments, given identical patient parameters. Often in clinical settings, the true patient dynamics and parameters are unknown. Thus, results from clinical data may not be repeatable under the same conditions as there could be unaccounted for variables in the methods.

3.5 Evaluation framework

To answer the given Research Questions 1 and 2, statistical metrics such as Mean Square Error and Mean Absolute Relative Error, and mathematical results (e.g. observability and computational complexity of estimators) will be used to utilized to quantify the results of the experiment such that the hypotheses can be tested.

In order to evaluate the estimation of patient-specific parameters, the methodology will follow a trajectory of investigating observability in insulin-glucose models, comparing state estimators, and then assessing parameter estimation methods. Each phase of the experiment will have their own respective patient cohort, and statistical metrics.

3.6 Observability of the Intensive Control Insulin-Nutrition-Glucose Model

In Chapter 3.3, a patient model was created for each individual patient based on the Intensive Control Insulin-Glucose model, with the ensemble of patient models being labeled as a patient cohort. For a particular patient model to be observable, it must be possible to infer the patient dynamics, given a sufficient number of observations. To establish if the formulation of the patient model is observable, a mathematical analysis can determine the relationship of the insulin-glucose dynamics and the observations as previously discussed in Section 2.1.3.

1000 patient models will be randomly sampled to investigate the observability of the Intensive Control Insulin-Nutrition-Glucose model across various permutations of each random nominal state vector. The maximum value of each state* x_{max} will be chosen as S

$$\begin{aligned} S &= \operatorname{diag}(x_{max}) \\ &= \operatorname{diag}(\begin{bmatrix} 17.5 & 260 & 220 & 11.5 & 57 \end{bmatrix}), \end{aligned} \tag{3.4}$$

*For this calculation, u_{en} and P are omitted. This is due to permutations of these values are directly represented by dynamics I and $P2$.

as well with a maximum perturbation constant of $c = 0.1$. With the following defined inputs, the observability gramian W_O can be derived to analyze the singular values of all the calculated matrices using single value decomposition. To also analyze which states are the most observable respectively to each other, the diagonals of the observability gramian can be used to propose that the diagonal entries that are greater in magnitude, those states provide more information in relation to the observations. Formulating an approximation of the observability will provide insight to the performance of the estimators in the following section.

3.7 Nonlinear State Estimation

The extended Kalman filter and the particle filter are two different nonlinear state estimation methods presented in Section 2.1. To compare each estimator with the Intensive Control Insulin-Nutrition-Glucose model, the computational complexity $\mathcal{C}$ of estimating all states over the total time and the mean squared error MSE of the blood glucose concentration estimates will be recorded for all patients and filters in the given virtual patient cohort. The simulations will span 12 hours to allow each estimator to converge, as well with a discrete time-step of 1 minute, with glucose measurements corrupted with additive Gaussian noise ($N(0, 0.25^2)$) sampled every 120 minutes. There will be iterations of this comparison with the following offsets of the initial states: $\pm 2\%, \pm 10\%, \pm 25\%$, with each iteration having a virtual cohort of 500 patients. Adding an offset to the initial state will gauge how each well estimator can adapt to various initial innovations in the estimate.

To ensure that model approximation errors can be modeled as a random process, artificial process noise can be added to the state estimates for both the extended Kalman filter and the particle filter. Due to the discrete time-step to measurement time-step ratio being 120:1, the process noise per sample was chosen as an expected amount of deviation between measurements divided by 120. The process noise w will be sampled at each discrete time-step

$$w \sim \mathcal{N}\bigg(0,\, \mathrm{diag}\left(5.0 \times 10^{-6},\, 1.0 \times 10^{-3},\, 1.0 \times 10^{-3},\, 5.0 \times 10^{-7},\, 2.5 \times 10^{-5},\, 1.0 \times 10^{-8}\right)\bigg). \tag{3.5}$$

The extended Kalman filter parameters will be defined as

1. Initial covariance matrix of $P_0 = \mathrm{diag}(1,\, 3,\, 3,\, 1,\, 2,\, 1)$

2. Process noise matrix of
$Q = \text{diag}\left(5.0\times10^{-6},\ 1.0\times10^{-3},\ 1.0\times10^{-3},\ 5.0\times10^{-7},\ 2.5\times10^{-5},\ 1.0\times10^{-8}\right)$
3. Measurement noise of $R = 0.25^2$

Particle filters containing $(100,\ 200,\ 300,\ 400,\ 500,\ 750,\ 1000,\ 1500,\ 2000)$ particles will be used to compare the relationship of the estimation performance over an increasing number of particles. The particle filter will be initialized by sampling particles from a multivariate Gaussian distribution with a mean of the initial state, and a covariance equal to the extended Kalman filter's Q.

To define the computational complexity of the estimators given the experimental setup, let τ_i denote the average total time to run each particle filter and the extended Kalman filter in seconds, with N being the total number of filters

$$\begin{aligned}\tau_{\text{min}} &= \min(\tau)\\ \mathcal{C}_i &= \frac{\tau_i}{\tau_{\text{min}}} \qquad i = 1,\ldots,N.\end{aligned} \tag{3.6}$$

The error of the estimators will be represented as the mean squared error MSE of the true blood glucose concentration sequence G and an estimated blood glucose concentration sequence $\hat{G}$ of length L

$$\text{MSE}_i = \frac{1}{L}\sum_{j=1}^{L}\left(G_i[j] - \hat{G}_i[j]\right)^2 \qquad i = 1,\ldots,K. \tag{3.7}$$

A Pareto curve can be utilized to visualize the "optimal" number of particles in terms of the trade off between computational complexity and error. The resulting number of particles needed per dimension will be calculated to gauge the rough number of particles needed to include additional parameters in the state-space in the following section.

3.8 Parameter Estimation

The Integral-based Parameter Identification method introduced in Section 2.2.2 will be the baseline implementation of estimating insulin sensitivity SI in patients and will be compared against the extended Kalman filter and the particle filter with state augmentation. Each method will try to estimate a constant insulin sensitivity and a piecewise insulin sensitivity that steps to a new value half-way through the experiment to draw conclusions on the limitations of each estimator. The simulation will last for 60 hours, as changes in S_I are relatively slow in time, with discrete time-steps of 1 minute. There will be 1000 total patients, and each filter will get an initial state with a positive 10% deviation from the true initial state. At 30 hours for the piecewise insulin S_I, a new value S_{I1} will be randomly sampled such the standard deviation is 14% of the current S_{I0}

$$S_{I1} \sim \exp\left(\mathcal{N}\left(\ln S_{I0},\, 0.14^2\right)\right). \tag{3.8}$$

Due to an additional state being included in the state space, the process noise in Equation (3.5) will be appended to include the uncertainty of S_I

$$w \sim \mathcal{N}\left(0,\, \mathrm{diag}\begin{pmatrix} 5.0\times10^{-6},\, 1.0\times10^{-3},\, 1.0\times10^{-3},\, 5.0\times10^{-7}, \\ 2.5\times10^{-5},\, 1.0\times10^{-8},\, 2.5\times10^{-8} \end{pmatrix}\right). \tag{3.9}$$

The extended Kalman filter is not well equipped to handle log normal dynamics. As a result, the process noise matrix Q has to differ slightly to compensate. The extended Kalman filter configuration for this section will be the following

1. Initial covariance matrix of $P_0 = \mathrm{diag}(1,\, 3,\, 3,\, 1,\, 2,\, 1, 10^{-9})$
2. Process noise matrix of
   $Q = \mathrm{diag}\left(5.0\times10^{-6},\, 1.0\times10^{-3},\, 1.0\times10^{-3},\, 5.0\times10^{-7},\, 2.5\times10^{-5},\, 1.0\times10^{-12}\right)$
3. Measurement noise of $R = 0.25^2$

The particle filter will contain the extrapolated "optimal" number of particles chosen from the previous section to accommodate for another dimension in the state-space, and an initial covariance matrix of $\mathrm{diag}\left(1,\, 3,\, 3,\, 1,\, 2,\, 1,\, 3\times10^{-2}\right)$. Assessing the estimation of the insulin sensitivity will be evaluated with the mean absolute relative difference MARD of the true insulin sensitivity S_I and the estimated insulin sensitivity $\hat{S}_I$

$$\mathrm{MARD}_i = \frac{1}{L}\sum_{j=1}^{L} \frac{\left| S_{Ii}[j] - \hat{S}_{Ii}[j] \right|}{S_{Ii}[j]} \quad i = 1, \ldots, K. \tag{3.10}$$

The mean absolute relative difference is chosen instead of the mean square error because insulin sensitivity is small in magnitude, thus, squaring error will induce difficult analyization of the overall performance.

In the next chapter, the results of the virtual patient cohort generation, observability, state estimation, and parameter estimation will be presented and analyzed.

Chapter 4

Results

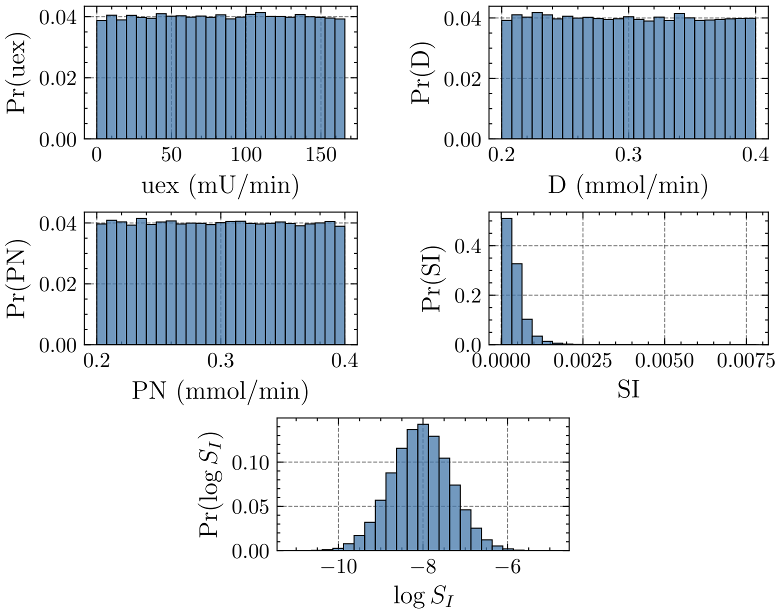
4.1 Generated Virtual Patient Cohort

Approximating probability mass functions was used to analyze the characteristics of the target population of critically-ill patients in the Karolinska University Hospital. Figure 4.1 is divided into two subfigures, with Figure 4.1a consisting of the probability mass functions of the inputs u_{ex}, D, and PN, as well as the insulin sensitivity parameter S_I, and Figure 4.1b is visualizing the probability mass functions of the derived initial dynamics G, I, Q, P_1, P_2, P, u_{en}.

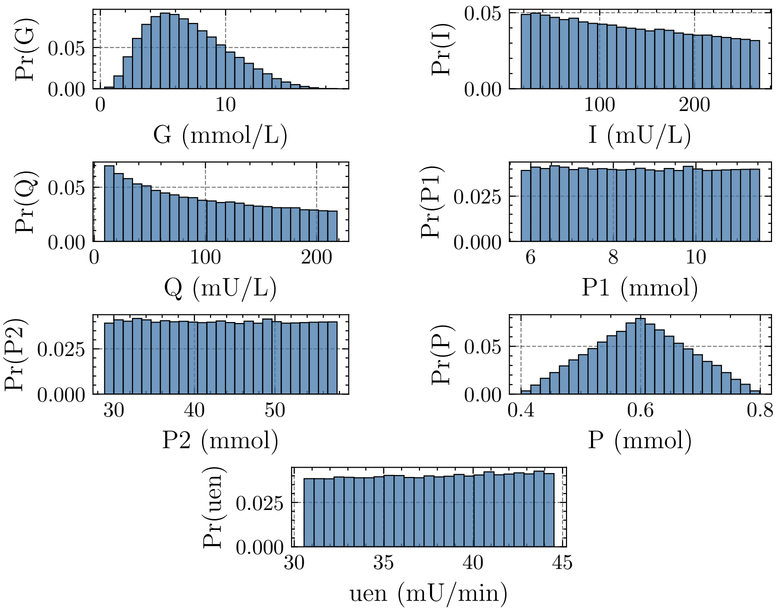
When numerous virtual patients were sampled, the figures present a diverse distribution of realistic initial dynamics from the randomly assigned inputs and insulin sensitivity. This allowed for clinically-validated sampling of many unique variations of critically-ill patient dynamics over time without relying on manual interventions. Previous work in virtual patient cohorts in glycaemic control was often limited to perturbations in previously recorded clinical data, but in this approach, a clear and reproducible distribution of patients was made feasible to analyze and compare the methods in the following sections.

4.2 Observability of the Intensive Control Insulin-Nutrition-Glucose Model

To investigate the observability of the Intensive Control Insulin-Nutrition-Glucose model, two methods were investigated. In Figure 4.2a, a probability mass function of eigenvalues from the 1000 calculated empirical observability



(a) Inputs and parameters



(b) Initial Dynamics

Figure 4.1: Probability mass functions of the generated virtual patient cohort characteristics.

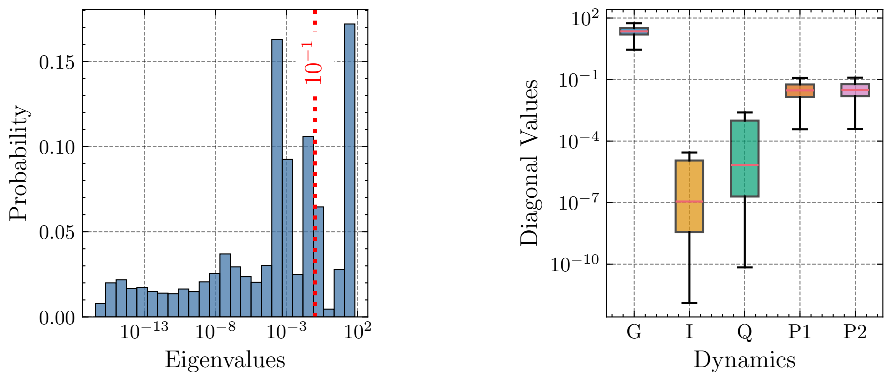
gramians were plotted. The eigenvalues are transformed in a log-scale to easier visualize the spread of values. The red dotted line denotes the threshold chosen to distinguish the eigenvalues that are considered observable (> 0.10) or unobservable (≤ 0.10). The eigenvalues range from 10^{-15} to 10^{2}. The percentage of eigenvalues considered observable is approximately 25.4%, which would suggest that, given all available measurements, about a quarter of the states provide insight into the observations. The other Figure 4.2b, a box plot for each dynamic in the empirical observability gramian is plotted to show the respective locality of the empirical observability matrix diagonal $\mathrm{diag}(W_O)$ over all patients. As a result of scaling all dynamics in the empirical observability gramian, the diagonal values are comparable amongst the dynamics.

The blood glucose concentration G is the most observable. This is to be expected as the observations in our state-space model are noisy blood glucose measurements. The blood glucose appearance in the stomach and the gut are the next observable states, with a median roughly of 2.95×10^{-2} and 3.07×10^{-2}, as well as an approximate interquartile range of 4.33×10^{-2} and 4.35×10^{-2} respectively. In Equation (2.1), the change in G is dependent on a constant consisting of a scaled multiple of the total glucose appearance P, which is a function of both the glucose appearance in the stomach and gut $P1$, $P2$. Thus, this representation can be attributed to why P_1 and P_2 are more observable compared to the plasma insulin I and the interstitial insulin Q, as the fraction $\frac{Q}{1+\alpha_G Q}$ is being multiplied by a relatively small value S_I, and changes in I take time to propagate to Equation (2.1).

Overall, the Intensive Control Insulin-Nutrition-Glucose model can not be classified as observable. For a system that is not observable, the performance of the state estimates may be hindered, as for each observation sampled, only the blood glucose and possibly the total glucose appearance can be uniquely identified. In practice, you could add more observations to increase the observability, but in the context of insulin-glucose dynamics, most institutions only have access to blood glucose measurements and not plasma insulin or C-peptide data. While poor observability is a limitation in state-space models, the following sections show that the model has capabilities, albeit imperfect, to estimate insulin-glucose dynamics.

4.3 Nonlinear State Estimation

In the first stage of comparing different nonlinear state estimators, each estimator is given an initial state that has a deviation in each state from the true



(a) Probability mass function of the eigenvalues

(b) Box plot of the diagonals across dynamics

Figure 4.2: Analysis across all calculated empirical observability gramians.

initial state. The notation of the extended Kalman filter is abbreviated as EKF and for a particle filter with N particles, it is represented as PF N. The mean squared error is averaged among all patients for each deviation in Table 4.1, as well as the standard deviation SD of the error, and an average mean squared error across all deviations is calculated. The extended Kalman filter overall performs the worst compared to the particle filters. When there are large deviations in the initial state of $\pm 25\%$, the mean squared error (0.265, 0.117) and the standard deviation (0.85, 0.55) are large respectively in comparison to a particle filter with only 100 particles (MSEs of 0.057 and 0.038, SDs of 0.17 and 0.12). Such improvements in performance with only using 100 particles, highlight the particle filter's ability to adapt quickly to errors in the state estimates when resampling after the initial measurement, compared to the extended Kalman filter that is dynamically propagating a point estimate of states. In the deviations of $\pm 2\%$ and $+10\%$, the extended Kalman filter outperforms or is comparable to the particle filters. It is a possibility that when resampling happens at the beginning of the simulation, regardless of the deviation, that the particle filter may be over-correcting because of the error till it adjusts after receiving another measurement.

To compare the overall performance of the filters, it is helpful to analyze Figure 4.3. The x-axis is the computational complexity of each estimator. The minimum time to run amongst all the estimators was the extended filter. Thus, its computational complexity is 1. The computational complexity of the rest of the estimators is the multiple of time to run relative to the extended Kalman filter. The y-axis is the average mean squared error for all deviations of the initial states, plotted in the log-domain. In terms of error, there is a plateau in

Table 4.1: MSE for different estimators and deviations of the initial state. Standard deviations of the MSE are shown in parentheses.

Estimator	**Deviations**						**Average**
	-25%	-10%	-2%	2%	10%	25%	
EKF	0.265	0.087	0.040	0.028	0.030	0.117	0.095
	(0.85)	(0.23)	(0.11)	(0.08)	(0.10)	(0.55)	
PF 100	0.057	0.031	0.035	0.037	0.035	0.038	0.039
	(0.17)	(0.08)	(0.09)	(0.10)	(0.10)	(0.12)	
PF 200	0.055	0.033	0.035	0.036	0.035	0.037	0.038
	(0.16)	(0.08)	(0.09)	(0.10)	(0.10)	(0.12)	
PF 300	0.051	0.033	0.035	0.035	0.035	0.036	0.037
	(0.14)	(0.08)	(0.09)	(0.10)	(0.10)	(0.12)	
PF 400	0.046	0.033	0.034	0.035	0.035	0.036	0.037
	(0.13)	(0.08)	(0.09)	(0.10)	(0.10)	(0.11)	
PF 500	0.046	0.033	0.034	0.035	0.035	0.036	0.037
	(0.13)	(0.08)	(0.09)	(0.09)	(0.10)	(0.11)	
PF 750	0.046	0.033	0.034	0.034	0.035	0.035	0.036
	(0.13)	(0.08)	(0.09)	(0.09)	(0.10)	(0.11)	
PF 1000	0.038	0.033	0.034	0.035	0.035	0.036	0.035
	(0.10)	(0.08)	(0.09)	(0.09)	(0.10)	(0.12)	
PF 1500	0.036	0.034	0.034	0.035	0.036	0.036	0.035
	(0.09)	(0.08)	(0.09)	(0.10)	(0.10)	(0.12)	
PF 2000	0.037	0.033	0.035	0.035	0.036	0.037	0.035
	(0.09)	(0.08)	(0.09)	(0.10)	(0.10)	(0.12)	

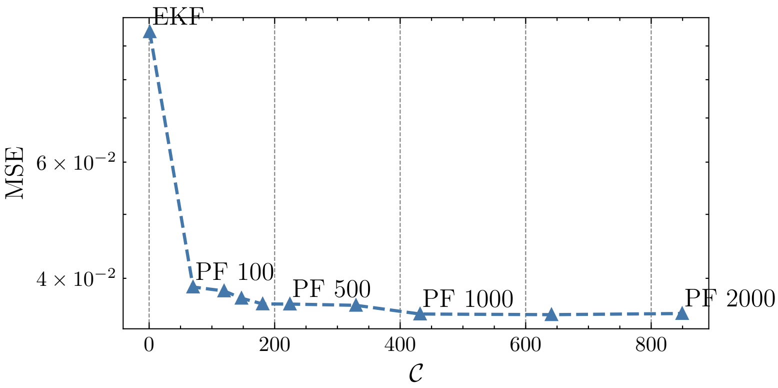


Figure 4.3: Comparison of the total average computational complexity and error across all estimators.

improvements in performance after 750 particles. The subjective "optimum" number of particles was chosen as 750 particles, as there is a balance of moderate performance and lower computational cost. It was assumed that for each dimension in the state space (total of 6), the number of particles needed P grows exponentially with a power of 2, which allows P to be estimated using the determined "optimal" number of particles

$$\begin{aligned} 750 &= P \times 6^2 \\ P &= \frac{750}{36} \\ P &\approx 20.8. \end{aligned} \tag{4.1}$$

For the next section, the insulin sensitivity will be included in the state-space. Consequently, the state space increases to 7. To have a rough range of the number of particles needed to accommodate a larger dimension in the state space, the approximate number of particles N was calculated as such

$$\begin{aligned} N &= P \times 7^2 \\ N &\approx 1020. \end{aligned} \tag{4.2}$$

The chosen "optimal" number of particles is much less computationally expensive compared to previous work [17] that used 5000 particles for a state-space with the same number of dimensions. Although the Aguirre et al. study did contain non-Gaussian measurement noise, the number of particles to estimate the posterior distribution with a known measurement model prior likely does not require that many particles. If we decided to choose 5000 particles, there would likely be little marginal gain in the performance and would result in 5 times the computational complexity. For a system that may require an estimator to run in approximately real-time, an estimator with 5000 particles would require lots of computational resources. Hence, our approach is more practical in clinical settings. If very limited computational resources are allotted, smaller sized particle filters or the extended Kalman filter can be utilized instead.

4.4 Parameter Estimation

State augmentation of insulin sensitivity in the extended Kalman filter and the particle filter was implemented to compare the estimation performance of the existing state estimators against the Integral-based Patient Identification method. Both constant and time-varying insulin-sensitivity were used to investigate how characteristics of the parameter affect estimation. Figures 4.4a and 4.4b display the performance of each method according to the time dependency of the parameter. MARD in these figures depicts the mean error for each time step, not the length of the sequence.

The Integral-based Parameter Identification method in both examples fluctuates between 0.05 and 0.07 MARD. The Integral-based Parameter Identification method can not account for measurement noise in the calculation. Therefore, the performance is partially dependent on the quality of the measurements. Any estimation of the insulin sensitivity from the Integral-based Parameter Identification method does not provide any information on the insulin sensitivity, as the output never converges towards the true value(s).

In Figure 4.4a, both the extended Kalman filter and the particle filter converge towards the true insulin sensitivity. The extended Kalman filter on average plateaus in the estimation of the insulin sensitivity around 0.03 MARD, while the particle filter with 1020 particles on average roughly converges to the true value. The extended Kalman filter often has higher variance in the estimate because particle filters average the variance over all the particles. Thus, the extended Kalman filter can also converge to the true value, but on average performs worse than the particle filter. In the time-varying parameter estimation case, when the insulin sensitivity randomly steps to a new value at 30 hours, it can be seen in Figure 4.4b, that the particle filter struggles to converge to the new true value compared to the extended Kalman filter. Before the insulin sensitivity steps to a new value at 1800 minutes, the particle filter roughly converged to true insulin sensitivity. As a result, many particles in the state-space were near the true value with most of the weight

Table 4.2: Comparison of the total average MARD in parameter estimation methods estimating insulin sensitivity

Method	**Constant** S_I	**Piecewise** S_I
EKF	0.035	0.046
PF 1020	0.027	0.062
Integral-based	0.063	0.061

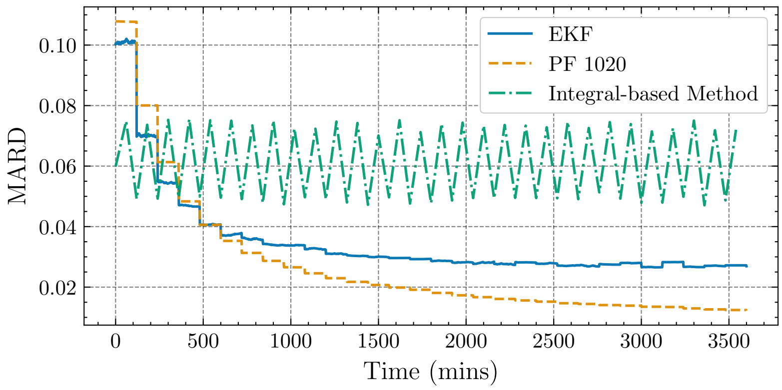
centered around those particles. When the insulin-sensitivity changes, there is sample impoverishment, and it takes a long time to propagate the particles throughout the state-space. While in the extended Kalman filter, the extended Kalman filter has higher variance in the covariance estimate P_k of the insulin-sensitivity and can slowly converge to the new true value. Further comments on how to mitigate the sample impoverishment in the particle filters will be discussed in the future work section in Chapter 5. To conclude, the overall performance of each method in both parameter estimation scenarios is shown in Table 4.2, with the particle filter with the lowest MARD in estimating constant S_I, and the extended Kalman filter performing the best in the time-varying S_I case.

4.5 Reliability Analysis

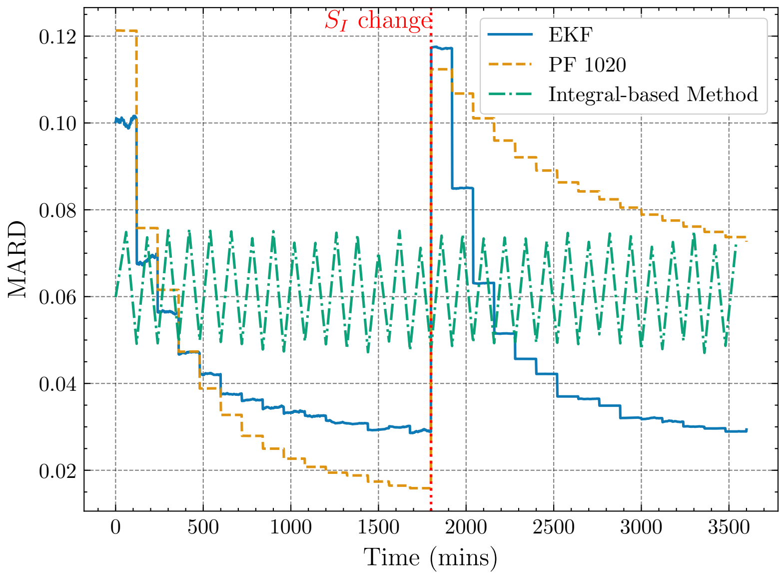
To ensure the reliability of each section, numerous virtual patients were used to best represent the target population of critically-ill patients in the Karolinska University Hospital. Having many simulations reduces the likelihood that a different permutation of the patient cohort will change the result of the experiment. All patient cohort characteristics and filter parameters are clearly documented in the report, and in the source code to ensure repeatability of the experiment, if someone else chooses to replicate the study.

4.6 Validity Analysis

In the patient cohort generation, the inputs and parameters are chosen from clinically validated values. To identify the initial patient dynamics, it is assumed that the patient is at steady-state. Assuming the patient is at steady-state is a valid assumption, as the patient generally do not have external nutrition right before the start of their stay. Given these assumptions, the



(a) Constant S_I



(b) Piecewise S_I

Figure 4.4: MARD over time in parameter estimation methods estimating insulin sensitivity

characteristics of the initial patient dynamics are also valid. When typically assessing observability in nonlinear systems, empirical methods are used as typical observability calculations are not tractable. Previous work in the field of glycaemic control and other fields used the empirical observability gramian as a tool to gauge how observable the dynamics are. While the exact values of the eigenvalues or the observability gramian are just approximations, they are valid as a guide to assess the behavior of the model of interest. Regarding state and parameter estimation, performance is typically measured on some error metric to gauge performance, which validates the choice of using mean squared error and mean absolute relative difference. Overall, all the design choices to answer the research questions are based on validated assumptions clinically or from previous work in state and parameter estimation.

Chapter 5

Conclusions

5.1 Limitations

5.2 Future work

5.2.1 What has been left undone?

5.2.1.1 Cost analysis

5.2.2 Next obvious things to be done

5.3 Reflections

References

[1] K. M. Dungan, S. S. Braithwaite, and J.-C. Preiser, "Stress hyperglycaemia," *The Lancet*, vol. 373, no. 9677, pp. 1798–1807, May 2009. doi: 10.1016/S0140-6736(09)60553-5. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0140673609605535 [Page 1.]

[2] P. Paolisso, A. Foà, L. Bergamaschi, F. Donati, M. Fabrizio, C. Chiti, F. Angeli, S. Toniolo, A. Stefanizzi, M. Armillotta, P. Rucci, G. Iannopollo, G. Casella, C. Marrozzini, N. Galiè, and C. Pizzi, "Hyperglycemia, inflammatory response and infarct size in obstructive acute myocardial infarction and MINOCA," *Cardiovascular Diabetology*, vol. 20, no. 1, p. 33, Feb. 2021. doi: 10.1186/s12933-021-01222-9. [Online]. Available: https://doi.org/10.1186/s12933-021-01222-9 [Page 1.]

[3] G. Song, X. Liu, Z. Lu, J. Guan, X. Chen, Y. Li, G. Liu, G. Wang, and F. Ma, "Relationship between stress hyperglycaemic ratio (SHR) and critical illness: a systematic review," *Cardiovascular Diabetology*, vol. 24, no. 1, p. 188, May 2025. doi: 10.1186/s12933-025-02751-3. [Online]. Available: https://doi.org/10.1186/s12933-025-02751-3 [Page 1.]

[4] C. H. Lang, C. Dobrescu, and K. Mészáros, "Insulin-mediated glucose uptake by individual tissues during sepsis," *Metabolism*, vol. 39, no. 10, pp. 1096–1107, Oct. 1990. doi: 10.1016/0026-0495(90)90172-9. [Online]. Available: https://www.sciencedirect.com/science/article/pii/0026049590901729 [Page 1.]

[5] R. C. ANDREWS and B. R. WALKER, "Glucocorticoids and insulin resistance: old hormones, new targets," *Clinical Science*, vol. 96, no. 5, pp. 513–523, Apr. 1999. doi: 10.1042/cs0960513. [Online]. Available: https://doi.org/10.1042/cs0960513 [Page 1.]

[6] R. E. Shangraw, F. Jahoor, H. Miyoshi, W. A. Neff, C. A. Stuart, D. N. Herndon, and R. R. Wolfe, "Differentiation between septic and postburn insulin resistance," *Metabolism*, vol. 38, no. 10, pp. 983–989, Oct. 1989. doi: 10.1016/0026-0495(89)90010-3. [Online]. Available: https://www.sciencedirect.com/science/article/pii/0026049589900103 [Page 1.]

[7] T. Heise, L. Heinemann, and A. A. R. Starke, "Simulated postaggression metabolism in healthy subjects: Metabolic changes and insulin resistance," *Metabolism*, vol. 47, no. 10, pp. 1263–1268, Oct. 1998. doi: 10.1016/S0026-0495(98)90334-1. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0026049598903341 [Page 1.]

[8] G. V. d. Berghe, P. Wouters, F. Weekers, C. Verwaest, F. Bruyninckx, M. Schetz, D. Vlasselaers, P. Ferdinande, P. Lauwers, and R. Bouillon, "Intensive Insulin Therapy in Critically Ill Patients," *New England Journal of Medicine*, vol. 345, no. 19, pp. 1359–1367, Nov. 2001. doi: 10.1056/NEJMoa011300 _eprint: https://www.nejm.org/doi/pdf/10.1056/NEJMoa011300. [Online]. Available: https://www.nejm.org/doi/full/10.1056/NEJMoa011300 [Page 1.]

[9] F. M. Brunkhorst, C. Engel, F. Bloos, A. Meier-Hellmann, M. Ragaller, N. Weiler, O. Moerer, M. Gruendling, M. Oppert, S. Grond, D. Olthoff, U. Jaschinski, S. John, R. Rossaint, T. Welte, M. Schaefer, P. Kern, E. Kuhnt, M. Kiehntopf, C. Hartog, C. Natanson, M. Loeffler, and K. Reinhart, "Intensive Insulin Therapy and Pentastarch Resuscitation in Severe Sepsis," *New England Journal of Medicine*, vol. 358, no. 2, pp. 125–139, Jan. 2008. doi: 10.1056/NEJMoa070716 _eprint: https://www.nejm.org/doi/pdf/10.1056/NEJMoa070716. [Online]. Available: https://www.nejm.org/doi/full/10.1056/NEJMoa070716 [Page 1.]

[10] NICE-SUGAR Study Investigators, S. Finfer, B. Liu, D. R. Chittock, R. Norton, J. A. Myburgh, C. McArthur, I. Mitchell, D. Foster, V. Dhingra, W. R. Henderson, J. J. Ronco, R. Bellomo, D. Cook, E. McDonald, P. Dodek, P. C. Hébert, D. K. Heyland, and B. G. Robinson, "Hypoglycemia and risk of death in critically ill patients," *The New England Journal of Medicine*, vol. 367, no. 12, pp. 1108–1118, Sep. 2012. doi: 10.1056/NEJMoa1204942 [Page 1.]

[11] L. Evans, A. Rhodes, W. Alhazzani, M. Antonelli, C. M. Coopersmith, C. French, F. R. Machado, L. Mcintyre, M. Ostermann, H. C. Prescott, C. Schorr, S. Simpson, W. J. Wiersinga, F. Alshamsi, D. C. Angus, Y. Arabi, L. Azevedo, R. Beale, G. Beilman, E. Belley-Cote, L. Burry, M. Cecconi, J. Centofanti, A. Coz Yataco, J. De Waele, R. P. Dellinger, K. Doi, B. Du, E. Estenssoro, R. Ferrer, C. Gomersall, C. Hodgson, M. H. Møller, T. Iwashyna, S. Jacob, R. Kleinpell, M. Klompas, Y. Koh, A. Kumar, A. Kwizera, S. Lobo, H. Masur, S. McGloughlin, S. Mehta, Y. Mehta, M. Mer, M. Nunnally, S. Oczkowski, T. Osborn, E. Papathanassoglou, A. Perner, M. Puskarich, J. Roberts, W. Schweickert, M. Seckel, J. Sevransky, C. L. Sprung, T. Welte, J. Zimmerman, and M. Levy, "Surviving sepsis campaign: international guidelines for management of sepsis and septic shock 2021," *Intensive Care Medicine*, vol. 47, no. 11, pp. 1181–1247, Nov. 2021. doi: 10.1007/s00134-021-06506-y. [Online]. Available: https://doi.org/10.1007/s00134-021-06506-y [Page 1.]

[12] J. G. Chase, G. M. Shaw, T. Lotz, A. LeCompte, J. Wong, J. Lin, T. Lonergan, M. Willacy, and C. E. Hann, "Model-Based Insulin and Nutrition Administration for Tight Glycaemic Control in Critical Care," *Current Drug Delivery*, vol. 4, no. 4, pp. 283–296. doi: 10.2174/156720107782151223. [Online]. Available: https://www.eurekaselect.com/article/4990 [Pages 2 and 5.]

[13] T. F. Lotz, J. G. Chase, K. A. McAuley, G. M. Shaw, X.-W. Wong, J. Lin, A. LeCompte, C. E. Hann, and J. I. Mann, "Monte Carlo analysis of a new model-based method for insulin sensitivity testing," *Computer Methods and Programs in Biomedicine*, vol. 89, no. 3, pp. 215–225, Mar. 2008. doi: 10.1016/j.cmpb.2007.03.007. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0169260707000582 [Pages 2 and 5.]

[14] J. Lin, N. N. Razak, C. G. Pretty, A. Le Compte, P. Docherty, J. D. Parente, G. M. Shaw, C. E. Hann, and J. Geoffrey Chase, "A physiological Intensive Control Insulin-Nutrition-Glucose (ICING) model validated in critically ill patients," *Computer Methods and Programs in Biomedicine*, vol. 102, no. 2, pp. 192–205, May 2011. doi: 10.1016/j.cmpb.2010.12.008. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0169260710003007 [Pages 2 and 5.]

[15] C. E. Hann, J. G. Chase, J. Lin, T. Lotz, C. V. Doran, and G. M. Shaw, "Integral-based parameter identification for long-term dynamic verification of a glucose–insulin system model," *Computer Methods and Programs in Biomedicine*, vol. 77, no. 3, pp. 259–270, Mar. 2005. doi: 10.1016/j.cmpb.2004.10.006. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0169260704002123 [Pages 2, 6, and 14.]

[16] A. L. Compte, J. G. Chase, A. Lynn, C. Hann, G. Shaw, X.-W. Wong, and J. Lin, "Blood Glucose Controller for Neonatal Intensive Care: Virtual Trials Development and First Clinical Trials," *Journal of Diabetes Science and Technology*, vol. 3, no. 5, pp. 1066–1081, Sep. 2009. doi: 10.1177/193229680900300510. [Online]. Available: https://journals.sagepub.com/doi/10.1177/193229680900300510 [Pages 2 and 6.]

[17] E. Aguirre-Zapata, J. Cardenas-Cartagena, and J. Garcia-Tirado, "Glycemic Monitoring in Critical Care using Nonlinear State Estimators," *IFAC-PapersOnLine*, vol. 50, no. 1, pp. 4430–4435, Jul. 2017. doi: 10.1016/j.ifacol.2017.08.369. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2405896317307127 [Pages 2, 12, and 30.]

[18] C. Eberle and C. Ament, "Identifiability and online estimation of diagnostic parameters with in the glucose insulin homeostasis," *Biosystems*, vol. 107, no. 3, pp. 135–141, Mar. 2012. doi: 10.1016/j.biosystems.2011.11.003. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0303264711001857 [Pages 2 and 12.]

[19] J. G. Chase, G. M. Shaw, J. Lin, C. V. Doran, C. Hann, M. B. Robertson, P. M. Browne, T. Lotz, G. C. Wake, and B. Broughton, "Adaptive bolus-based targeted glucose regulation of hyperglycaemia in critical care," *Medical Engineering & Physics*, vol. 27, no. 1, pp. 1–11, Jan. 2005. doi: 10.1016/j.medengphy.2004.08.006. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1350453304001420 [Page 6.]

[20] S. Särkkä and L. Svensson, "Bayesian Filtering and Smoothing." [Pages 8, 9, 10, and 14.]

[21] J. S. Liu, *Monte Carlo Strategies in Scientific Computing*, ser. Springer Series in Statistics. New York, NY: Springer, 2004. ISBN 978-0-387-76369-9 978-0-387-76371-2. [Online]. Available: http://link.springer.com/10.1007/978-0-387-76371-2 [Page 10.]

[22] C. Kuptametee and N. Aunsri, "A review of resampling techniques in particle filtering framework," *Measurement*, vol. 193, p. 110836, Apr. 2022. doi: 10.1016/j.measurement.2022.110836. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0263224122001312 [Page 10.]

[23] J. Hahn and T. F. Edgar, "An improved method for nonlinear model reduction using balancing of empirical gramians," *Computers & Chemical Engineering*, vol. 26, no. 10, pp. 1379–1397, Oct. 2002. doi: 10.1016/S0098-1354(02)00120-5. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0098135402001205 [Page 11.]

[24] S. Davidson, C. Pretty, V. Uyttendaele, J. Knopp, T. Desaive, and J. G. Chase, "Multi-input stochastic prediction of insulin sensitivity for tight glycaemic control using insulin sensitivity and blood glucose data," *Computer Methods and Programs in Biomedicine*, vol. 182, p. 105043, Dec. 2019. doi: 10.1016/j.cmpb.2019.105043. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0169260718314342 [Page 18.]

[25] J. Dormand and P. Prince, "A family of embedded Runge-Kutta formulae," *Journal of Computational and Applied Mathematics*, vol. 6, no. 1, pp. 19–26, Mar. 1980. doi: 10.1016/0771-050X(80)90013-3. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/0771050X80900133 [Page 19.]

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